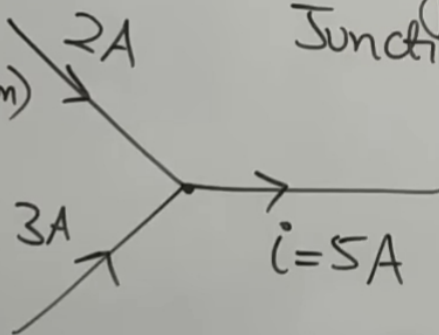
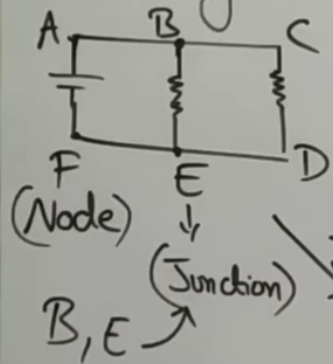


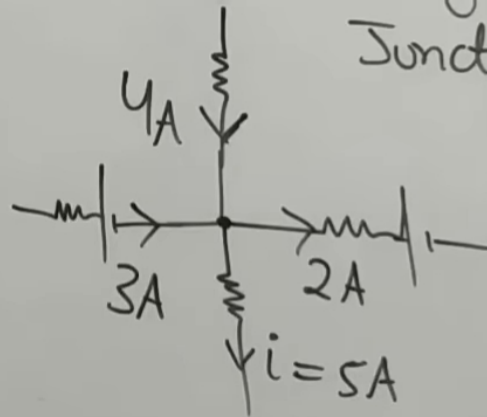
Kirchhoff's First Law / Junction Rule / Kirchhoff's Current Law (KCL)

At any Junction, $\rightarrow$ where 3 or more wire/conductor meets.



the sum of current entering the Junction

= Sum of current Leaving the Junction



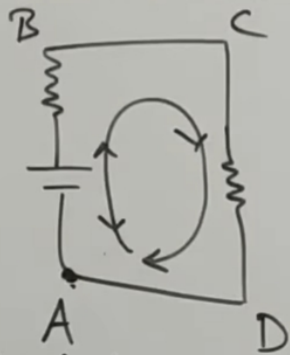
$$3 + 4 = 2 + i$$

$$7 = 2 + i$$

$$\boxed{i = 5A}$$

Kirchhoff's 2nd Law / Loop Rule / Kirchhoff's Voltage

Law (KVL) "The algebraic sum of change in potential around a CLOSED Loop is Zero."



Closed Loop
A B C D A

$$V_A - V_A = 0$$

Principle of Conservation
of Energy.

$$\sum \Delta V = 0 \quad \text{Closed Loop.}$$

Potential gain $\rightarrow +V$

Potential drop $\rightarrow -V$

$$\text{Closed Loop } \oint \pm V_1 \pm V_2 \pm V_3 \pm \dots \pm V_n = 0$$

KCL(Kirchoff's Current Law) is based on "**Conservation of Charge**"



Charge can neither be created nor be destroyed

Current is basically Charge per unit time

$$\mathbf{I = Q / t}$$

KVL (Kirchoff's Voltage Law) is based on "**Conservation of Energy**"

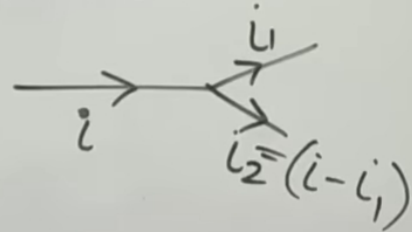
Basically, $\mathbf{V = \frac{W_{ext}}{q}}$

Work - joules / second

Thus, energy can neither be created nor be destroyed

How To Apply Kirchhoff's Law.

- ① Draw current from one/more cell $\rightarrow$ Svaradanu.
follow KCL for distribution of current.

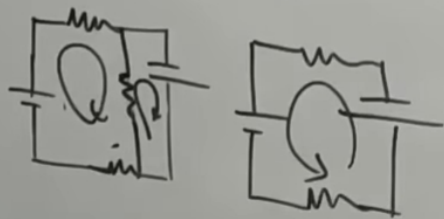


- ② Choosing a Loop.

i) Should be closed loop/loops.

Can involve any number of cells, resistors, capacitors etc.

- ③ Direction of Loop $\rightarrow$ path.

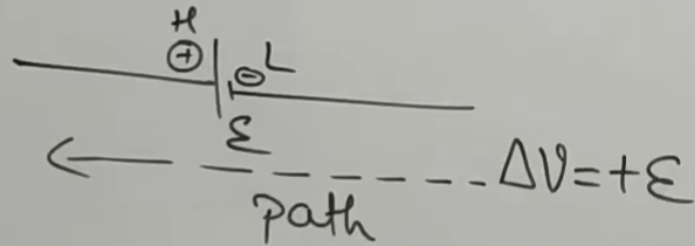
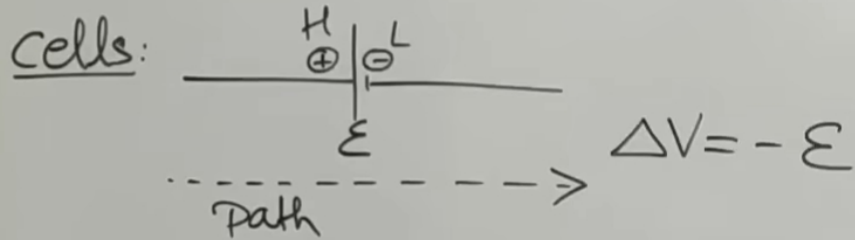


Clockwise / Anticlockwise.

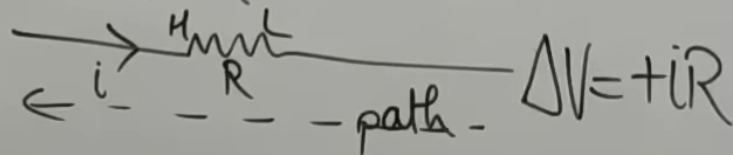
2 loops in same questions $\rightarrow$ take CW or ACW
one / both

How To Apply Kirchhoff's Law.

④ Potential drop or Potential gain.



Resistor.



Higher Potential $\longrightarrow$ Lower Potential
Potential drop

$$\Delta V = -ve$$

H $\longrightarrow$ L

Lower Potential $\longrightarrow$ Higher Potential
Potential gain

$$\Delta V = +ve$$

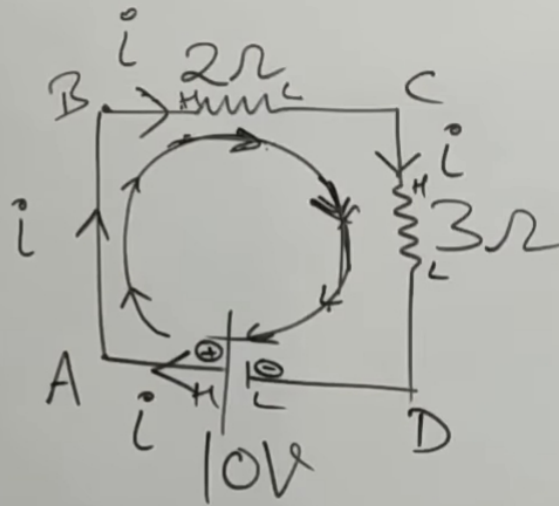
L $\longrightarrow$ H

***** Remember for KVL *****

In Cell, If Path follows from - to +, Take $\Delta v = +ve$

In resistors, If Current (i) and path are in opposite direction,
take $\Delta V = +ve$

Q1)



Loop ABCDA (Clockwise)

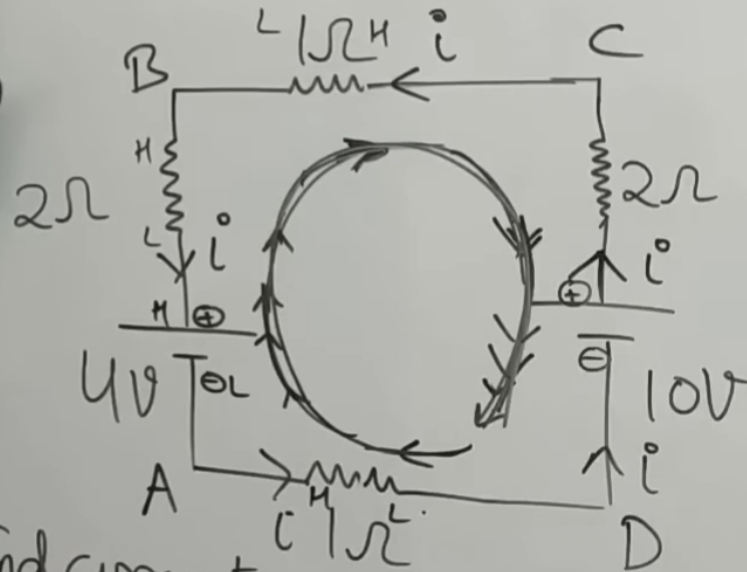
$$\sum \Delta V = 0$$

$$+10V - 2i - 3i = 0$$

$$5i = 10$$

$$i = \frac{10}{5} = 2A$$

Q2)



Find current
in each resistor.

Loop ABCDA (Clockwise.)

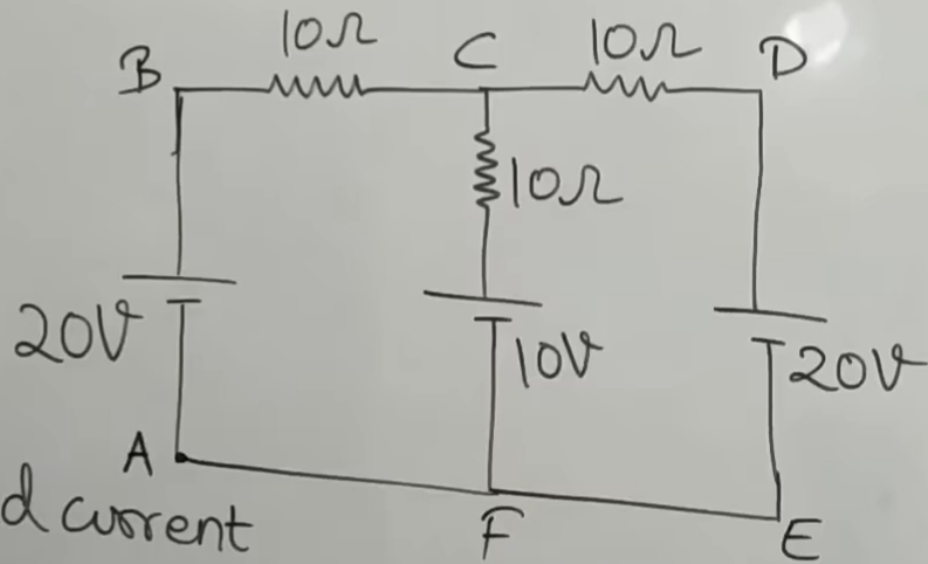
$$\sum \Delta V = 0$$

$$+4V + 2i + i + 2i - 10V + i = 0$$

$$6i = 6$$

$$\boxed{i = 1A}$$

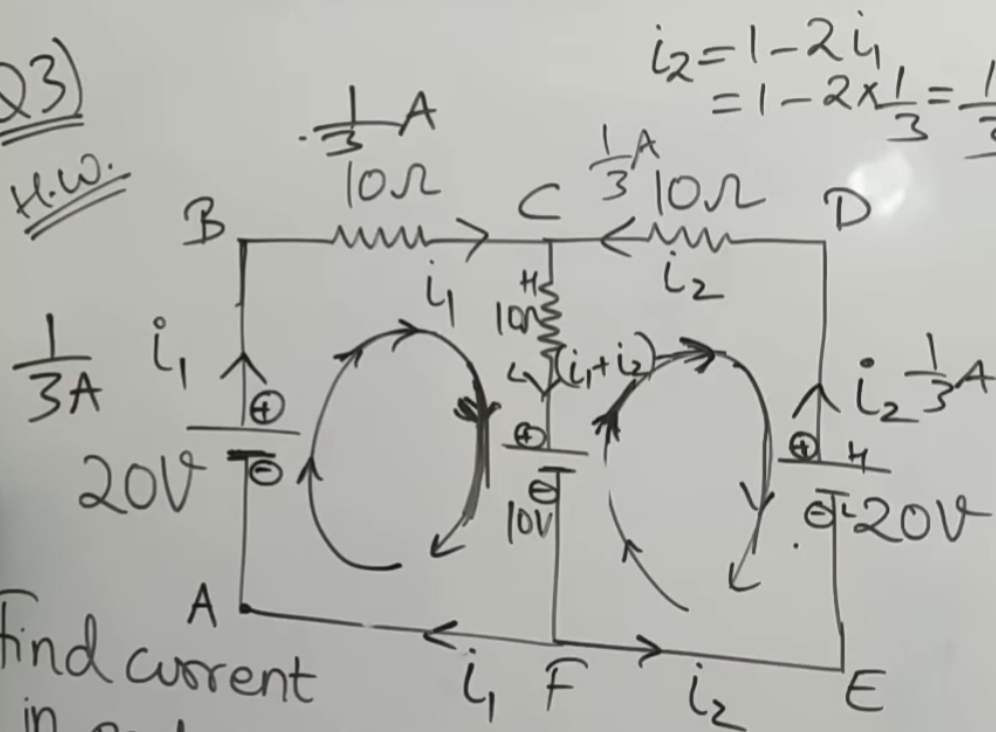
Q3)



Find current
in each resistor.

Q3)

H.W.



Find current
in each resistor.

Loop ABCFA

$$+20 - 10i_1 - 10(i_1 + i_2) - 10 = 0$$

$$2i_2 + i_1 = 1$$

$$3i_1 + 0 = 1$$

$$i_2 = 1 - 2i_1$$

$$= 1 - 2 \times \frac{1}{3} = \frac{1}{3} \text{ A}$$

$$20i_1 + 10i_2 = 10$$

$$2i_1 + i_2 = 1 \text{ --- (i)}$$

Loop CDEFC

$$+10 + 10(i_1 + i_2)$$

$$+10i_2 - 20 = 0$$

$$20i_2 + 10i_1 = 10$$

$$2i_2 + i_1 = 1 \text{ --- (ii)}$$

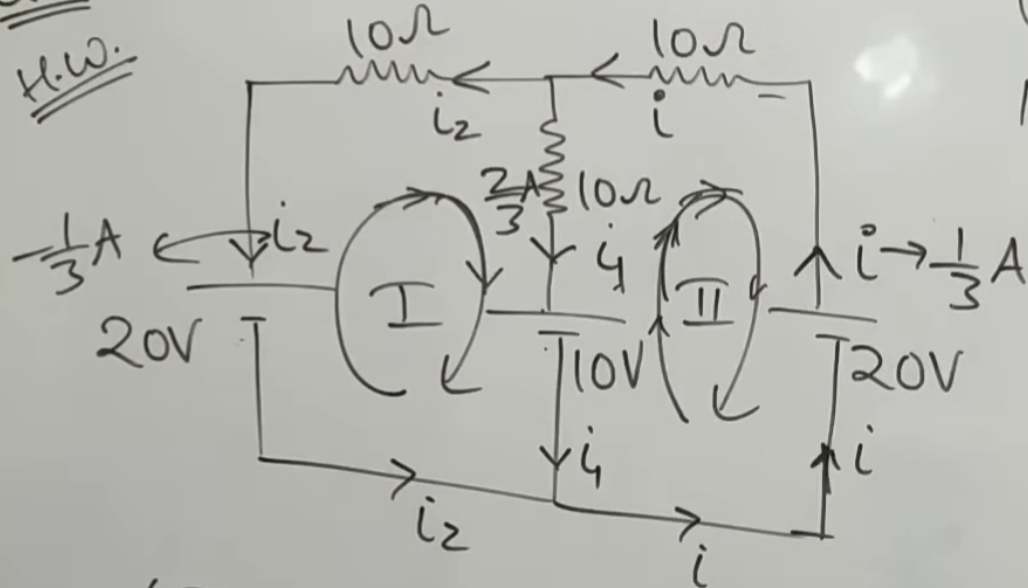
$$2i_1 + i_2 = 1 \times 2$$

$$3i_1 = 1$$

$$i_1 = \frac{1}{3}$$

Q3) $i = i_1 + i_2 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3} A$

H.W.



(I)

$$+20 + 10i_2 - 10i_1 - 10 = 0$$

$$10i_1 - 10i_2 = 10$$

$$\boxed{i_1 - i_2 = 1}$$

$$\boxed{\begin{aligned} \frac{2}{3} - i_2 &= 1 \\ i_2 &= \frac{2}{3} - 1 = -\frac{1}{3} A \end{aligned}}$$

(II)

$$10 + 10i_1 + 10i - 20 = 0$$

$$10i_1 + 10i = 10$$

$$i_1 + i = 1$$

$$i_1 + i_1 + i_2 = 1$$

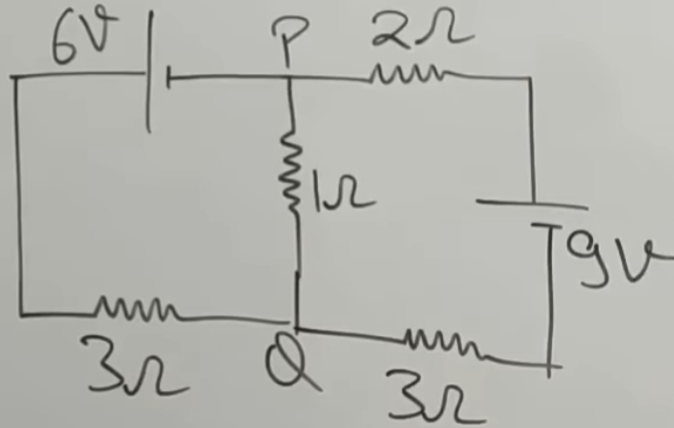
$$2i_1 + i_2 = 1$$

$$\textcircled{+} \quad i_1 - i_2 = 1$$

$$\hline 3i_1 = 2$$

$$\boxed{i_1 = \frac{2}{3} A}$$

Q4) IIT 2015
Mains



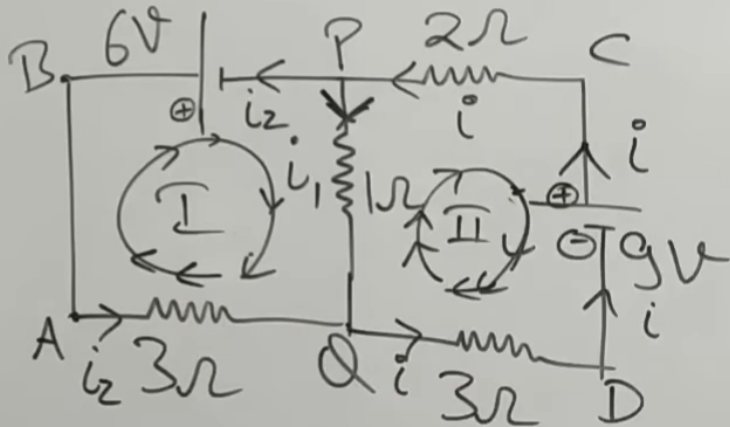
Find the current in
 1Ω resistor.

- a) $1.13\text{ A } P \rightarrow Q$
- b) $1.13\text{ A } Q \rightarrow P$
- c) $0.13\text{ A } Q \rightarrow P$
- d) $0.13\text{ A } P \rightarrow Q$

Q4) IIT 2015 Mains

$$i_1 = -\frac{3}{24} = -\frac{1}{8} = -0.125 \text{ A} \approx -0.13 \text{ A}$$

Loop ABPQA



$$-6 - i_1 + 3i_2 = 0$$

$$3i_2 - i_1 = 6 \quad \text{--- (i)}$$

Loop PCDQP

$$-9 + 3i + i_1 + 2i = 0$$

$$5i + i_1 = 9$$

$$5(i_1 + i_2) + i_1 = 9$$

$$5i_1 + 5i_2 + i_1 = 9$$

$$6i_1 + 5i_2 = 9 \quad \text{--- (ii)}$$

Find the current in 1Ω resistor.

$$i = i_1 + i_2$$

$$3i_2 - i_1 = 6 \times 5$$

$$6i_1 + 5i_2 = 9 \times 3$$

$$-23i_1 = 3$$

$$i_1 = -\frac{3}{23}$$

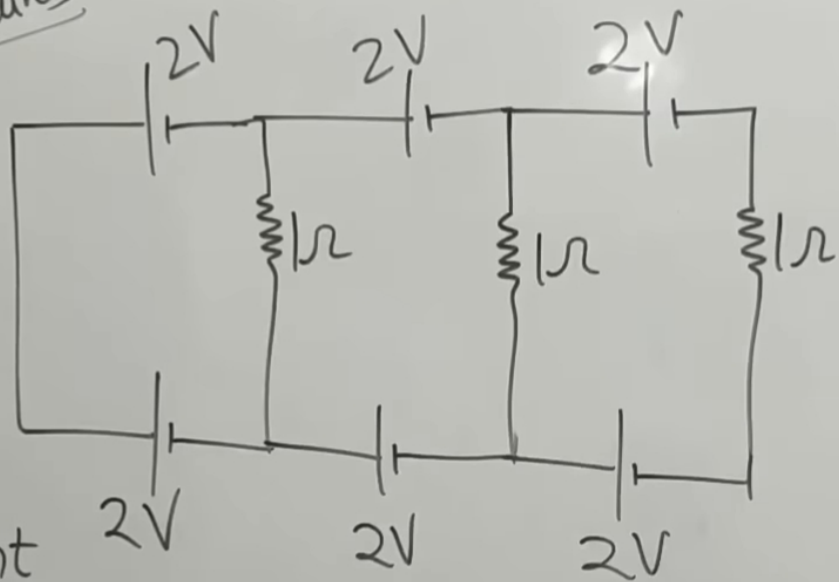
$$a) 1.13 \text{ A } P \rightarrow Q$$

$$b) 1.13 \text{ A } Q \rightarrow P$$

$$c) 0.13 \text{ A } Q \rightarrow P$$

$$d) 0.13 \text{ A } P \rightarrow Q$$

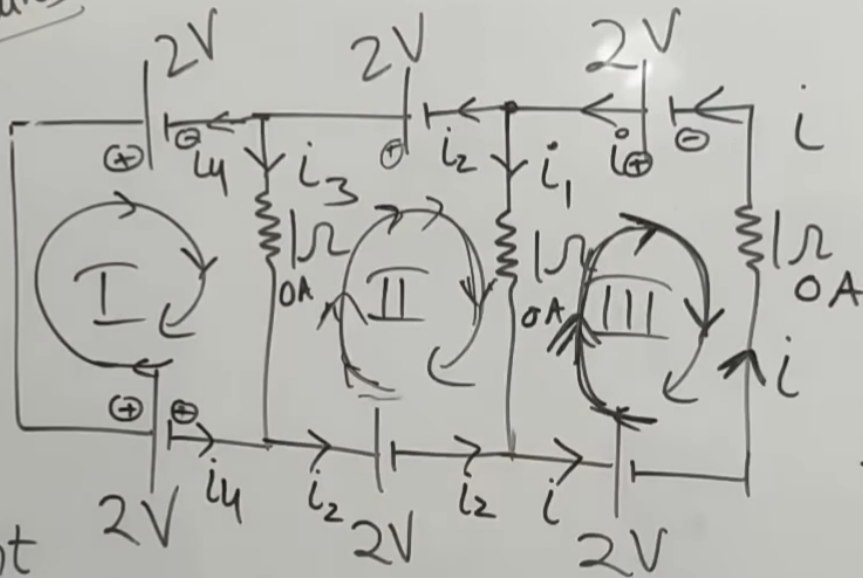
Q6) IIT JEE
2017 Mains



Find current
in each resistor

- a) 0.25A
- b) 0.5A
- c) 0A
- d) 0.75A.

Q6) IIT JEE
2017 Mains



Find current
in each resistor

- a) 0.25A
- b) 0.5A
- ☒ c) 0A
- d) 0.75A.

Loop I

$$+2V - 2V - i_3 \times 1 = 0$$

$$\boxed{i_3 = 0}$$

Loop II

$$+2V + i_3 - 2V - i_1 = 0$$

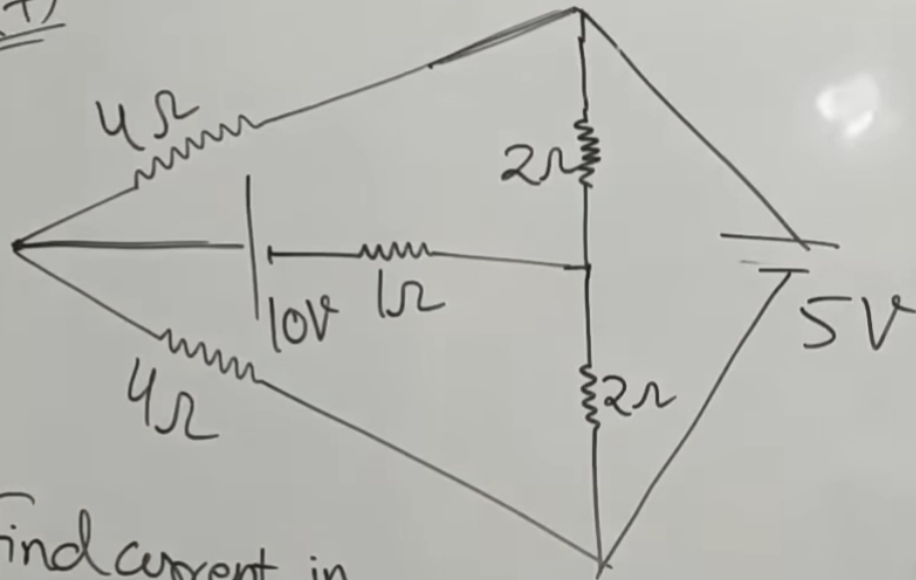
$$\boxed{i_3 = i_1 = 0}$$

Loop III

$$+2V + i_1 - 2V + i = 0$$

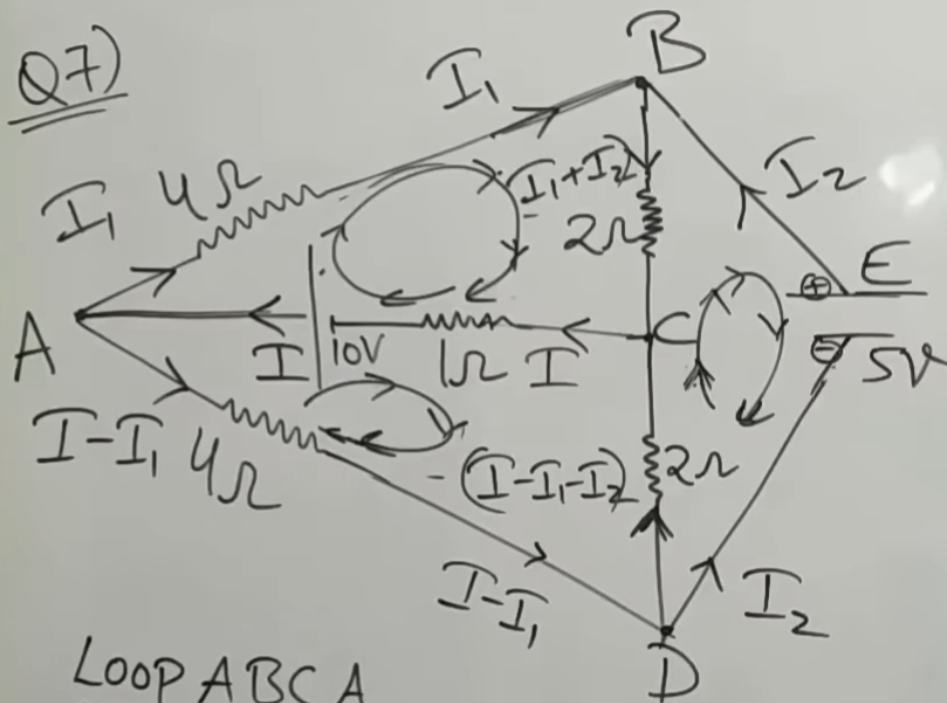
$$i = -i_1 = 0$$

Q7)



Find current in
each resistor.

Q7)



Loop ABCA

$$+10 - 4I_1 - 2(I_1 + I_2) - I = 0$$

$$\underline{I + 6I_1 + 2I_2 = 10 \text{ (i)}}$$

Loop ACDA

$$-10 + I + 2(I - I_1 - I_2) + 4(I - I_1) = 0$$

$$\underline{7I - 6I_1 - 2I_2 = 10 \text{ (ii)}}$$

Loop III BEDB

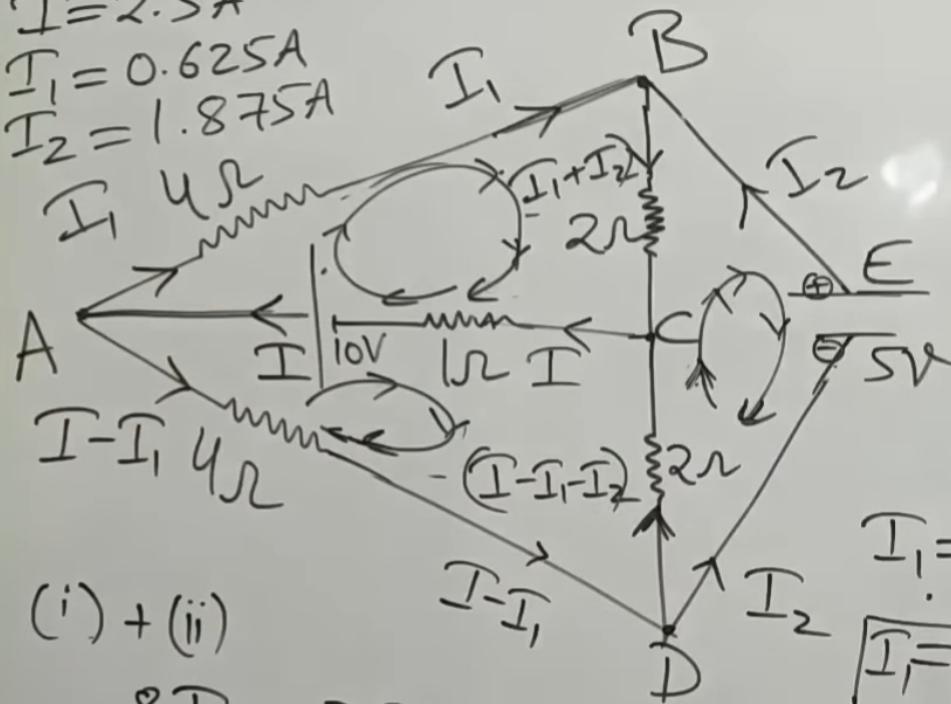
$$-5 - 2(I - I_1 - I_2) + 2(I_1 + I_2) = 0$$

$$\underline{4I_1 + 4I_2 - 2I = 5 \text{ (iii)}}$$

$$I = 2.5A$$

$$I_1 = 0.625A$$

$$I_2 = 1.875A$$



$$6I_1 + 2I_2 = 7.5 \times 2$$

$$\ominus \frac{4I_1 + 4I_2 = 10}{8I_1 + 0 = 5}$$

$$\underline{7I - 6I_1 - 2I_2 = 10 \text{ (ii)}}$$

$$(i) + (ii)$$

$$8I = 20$$

$$I = \frac{20}{8} = \frac{5}{2} = 2.5A$$

$$\underline{I + 6I_1 + 2I_2 = 10 \text{ (i)}}$$

$$I_1 = \frac{5}{8}$$

$$\boxed{I_1 = 0.625A}$$

$$4I_1 + 4I_2 - 2I = 5 \text{ (iii)}$$

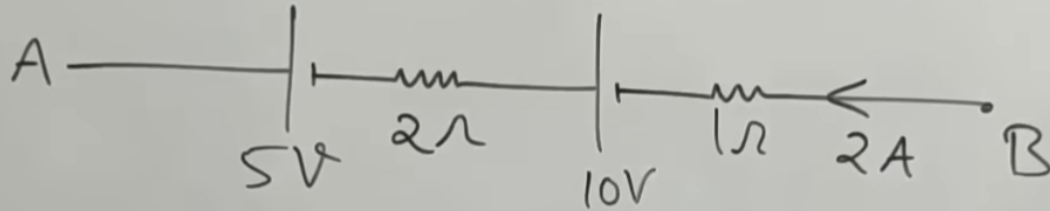
$$3 \times \frac{5}{8} + 2I_2 = \frac{15}{2}$$

$$2I_2 = \frac{15}{2} - \frac{15}{4}$$

$$\boxed{I_2 = \frac{15}{8}}$$

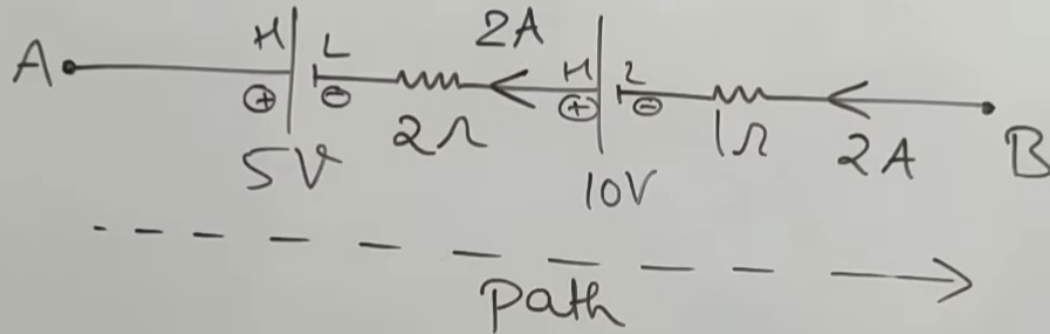
Potential difference between 2 points.

$$V_A - V_B = ?$$



Potential difference between 2 points.

$$V_A - V_B = 8V.$$

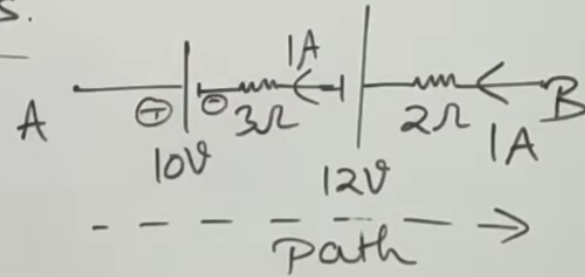


$$V_A - 5 + 2 \times 2 - 10 + 1 \times 2 = V_B$$

$$V_A - 9 = V_B$$

$$V_A - V_B = 9V$$

Potential difference between 2 points.

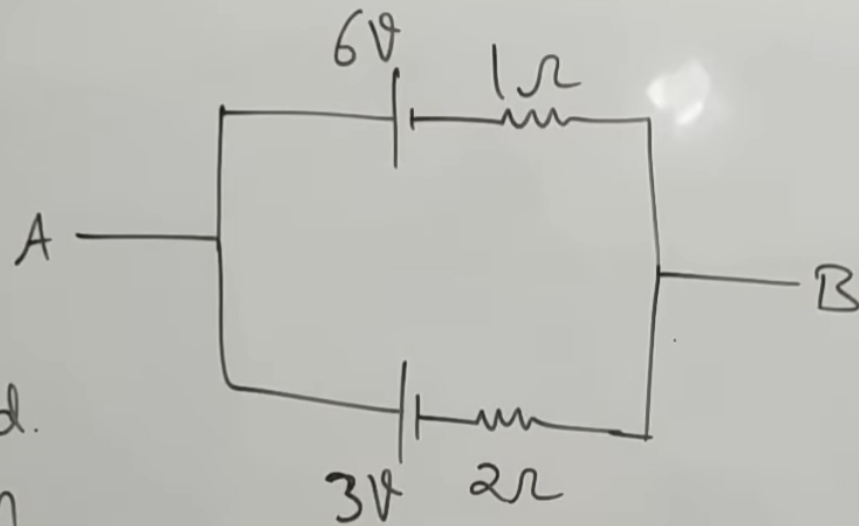


$$V_A - 10 + 3 \times 1 + 12 + 2 \times 1 = V_B$$

$$V_A + 7 = V_B$$

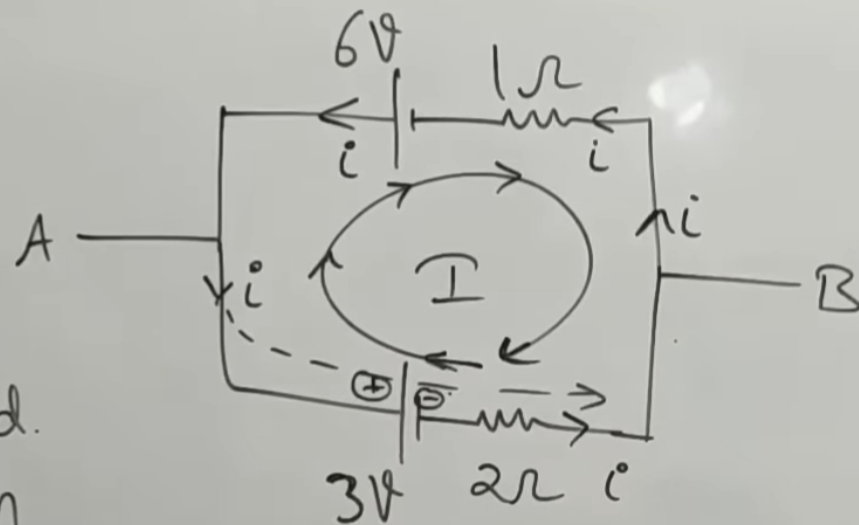
$$\boxed{V_A - V_B = -7V}$$

11T
2011



Find p.d.
between
A & B.

11T
2011



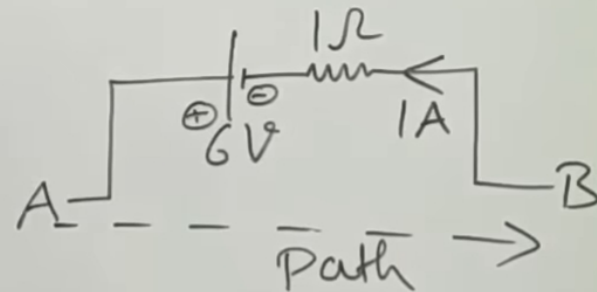
Find p.d.
between
A & B.

$$V_A - 3 - 2 \times i = V_B$$

$$3V - 6V + i \times 1 + 2 \times i = 0$$

$$-3 = -3i$$

$$i = 1A$$



$$V_A - 6 + 1 \times 1 = V_B$$

$$V_A - 5 = V_B$$

$$V_A - V_B = 5V$$